

Optimized VGG16 Model for Advanced Classification of Cotton Leaf Diseases

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Abstract— The cotton leaf disease inflection anywhere puts the production of agricultural products at risk as well as sustainable farming causing both problems and solutions of the meaningless disease detection. In this research, we propose an optimized Visual Geometry Group (VGG) 16 model for the advanced classification of four major cotton leaf diseases: leaf curl, bacterial blight, leaf spot, and wilt. Integration of deep learning methods, transfer learning as well and other approaches is what the model focuses on to raise the disease detection accuracy rate and realize the management of agriculture precision. And, there was a complete dataset, which included images of cotton plants from infected leaves with various diseases. This dataset was annotated properly and preprocessed. VGG16 model was pre-trained with a substantial data set of ImageNet image database and then fine-tuned to perform the cotton leaf disease classification task using existing transfer learning techniques. The achieved model efficacy has been evaluated using commonly accepted evaluation metrics. These include accuracy, precision, recall, and F1-score. Results show that the invented technique performs well when the revised VGG 16 model obtains satisfactory accuracy and disease-related metrics when applied to all three disease categories. The level of comparative analysis with current best models justifies done method efficacy and its strength in cotton leaf disease classification. This study's findings have an essential consequence on precision farming. It grants farmers a scalable solution relevant for the quick and automated monitoring, early diagnosis, and possibly treatment of plant diseases. Future investigation may rule in the superiority of applying the modified model to practical agricultural fields for ready accessibility and usability. Ultimately, these findings are moving us into the phase of development of agricultural informatics and offer evidence of the ability of deep learning to radically change approaches to fighting crop disease.

Keywords— Cotton leaf diseases, Disease classification, Deep learning, VGG16 model, Transfer learning, Precision agriculture, Crop disease management, Agricultural informatics, Image-based diagnosis, Machine learning.

I. INTRODUCTION

The global textile industry uses cotton (*Gossypium* family) as the main component of natural fibers forming the base for the production of fabrics [1]. Cotton occupies a yet prominent position in the agriculture of many countries for its substantial contribution to the world economy [2]. Nevertheless, cotton production coincides with several problems, the cotton pests it tells being the main ones that hinder the quality and quantity produced. One of the groups of diseases that may trigger cotton plants' leaves is cotton leaf spot, possibly bacterial blight, leaf spot, and wilt which can result in serious problems in agriculture and even retard agricultural sustainability [3]. Correct cotton leaf diseases entail proper and prompt disease management to get the exact diagnosis so that effective therapy will be efficient [4]. Morphologically, etiological analysis in agriculture has predominantly involved manual observations that were hinged on experts' instincts [3] [5]. This has come with numerous drawbacks that include delay in actual diagnosis and the need for prudence in arriving at meaningful interpretations since there exist possibilities of inaccuracies. Besides, owing to the growing hybridity and ascendancy of plant diseases, there is a high urgency for computer-aided rapid and straightforward solutions for precise detection and diagnosis [6]. In recent years, machine learning and computer vision technology have achieved a remarkable breakthrough in agricultural disease diagnosis, the autobiographic classification of plant disease with efficiency and scale as key benefits [7]. Deep learning, which belongs to the branches of machine learning, has recently been seen as a powerful image analysis tool that can deliver an extraordinary result in many areas like medical imaging, object recognition, text understanding, etc. In the agritech, deep learning models have been proven to be very useful for visual recognition, classification, and diagnosis of plant diseases based on images shared digitally [8]. Convolutional Neural Networks (CNNs) which have been most widely used to solve image classification problems, are capable of finding patterns in the way they transform raw pixel data into high-level features. The most popular CNN architecture that CNN

commonly employs for image classification is the one invented by the VGG, the VGG16 model in particular [9]. VGG16 receives accolades for simplicity and performance due to its simple configuration of several convolutional layers before they are followed by fully connected layers and ending with a softmax classifier used for multi-class classification. In this research paper, we propose an optimized VGG16 model for the advanced classification of cotton leaf diseases, focusing on four major leaf diseases prevalent in cotton cultivation: cotton leaf curl, stem rot, bacterial blight, leaf spot, and wilt. Our task will be to prove that the discriminating power of deep learning is the right choice for this particular problem by precisely distinguishing and recognizing these diseases in the cotton leaves with the help of high-resolution images [10]. Our research intends to utilize deep learning and also to tap into the anatomy of the cotton leaf which lies in signs of diseases. The main aim is to contribute towards developing automated, non-invasive, and scalable approaches to early disease detection and precise agriculture management in cotton cultivation.

II. LITERATURE REVIEW

Cotton farming has been challenged by many abiotic and biotic factors [1]. Diseases are among the most significant among these factors as they are adversely affecting the yield and quality of the cotton harvest [2]. The last few years have witnessed numerous researchers in the area of machine learning, deep-down learning models specifically, who have been majorly probing the complex aspect of diagnostics and control in agricultural diseases [5] [7]. This part starts with a review of the main articles addressing deep learning for plant disease classification and with more attention to those methodologies and approaches that can be used for cotton disease detection. The decision-making system using deep learning in plant disease classification based on several crops has been proven to affect the diseases of wheat, rice, maize, and soybeans [6] [11]. The methods were born on the principle of CNN, the class of deep learning models, which can be taught how to recognize deep objects in an automatic way looking only at the raw pixel data. CNN designs such as AlexNet, VGG, and ResNet have been tweaked and tailored for image-based plant disease detection. Inception, an alternative, has also been used [1]. It is worth noting the in-depth research on the role of deep learning models for automated diagnoses in the area of cotton disease classification is focused on [2]. The researchers have, among other things, created a CNN-based wilt detection model for cotton plants via hyperspectral imaging data. Their framework has shown the ability to be highly accurate in the differentiation of healthy and diseased plants which means the possibility of using deep learning could be great for cotton disease detection [4]. Likewise, the idea of using a deep learning approach to generate an automatic identification of cotton leaf diseases having leaf images as inputs has also come up by many writers. It used a ResNet-18 Network Architecture (ResNet) with transfer learning techniques in classifying five widespread cotton leaf diseases including Fusarium wilt, Verticillium wilt, and cotton curl leaf [10]. The results showed that their model was a great improvement from the conventional machine learning approaches [4]. This goes to demonstrate the superiority of deep learning in cotton disease diagnosis [9]. However, these researches succeeded in the classification of cotton diseases

but there is a need for more complex models that can separate a large range of the diseases and recognize them in a very accurate manner [12]. Similarly and most intimately, feeding deep learning models with typical real-world agricultural data exposes challenges such as data collection, model generalization, and scalability. Lately, in the deep learning area a variety of research has been carried out primarily but not limited to model optimization and transfer learning, these developments give rise to improving the disease classification model accuracy and robustness [6]. Transfer learning enables learning knowledge from one job to another and has been broadly used to deal with losses of information learning from scarcely annotated agricultural series [7]. Visual of specific cotton diseases classifying domain, fewer investigations can cover the development of specialized deep learning models for advanced disease classification. This article will be based on the gap which is the lack of a VGG16 model optimized for cotton leaf disease categorization to be overcome by proposing a specifically for this task [10]. Through transfer learning and fine-tuning techniques, we strive to increase the rate of disease diagnosis in cotton farming with the prospect of developing more precise strategies for disease management and crop protection [8].

III. METHODOLOGY

A. Data Collection and Preparation:

This research involves an overall dataset of cotton leaves from a different source to guarantee a variety of cotton leaf diseases and representation [12]. The dataset comprises high-resolution images captured in an environment with controlled lighting and background. The dataset includes images of four major cotton leaf diseases: cotton leaf curl, bacterial blight, leaf spot, and wilt. Every disease entity is demonstrated with a noteworthy number of images to allow the fine model training and its evaluation. Table I shows the distribution of the dataset across the disease categories [13].

TABLE I. DATASET DISTRIBUTION

Disease classes	Number of Images
Cotton Leaf Curl	500
Bacterial Blight	450
Leaf Spot	400
Wilt	350
Total	1700

B. Preprocessing:

The deep learning model that was given the images as casually as it was before it was fed into the preprocessing techniques to improve the quality and congruity of the dataset. This includes:

C. Resizing:

All images were resampled to uniform resolution (for example, 224x224 pixels) for the modeling model of VGG16 so that all samples had the same size [11].

D. Normalization:

Only Pixel values of the images were used, after being scaled to a 0-1 interval. This setting was made during the model training, to ensure proper convergence [14].

E. Augmentation:

As for the data augmentation methods utilized, these include rotation, flipping, and random cropping mechanisms to provide more variations and improve the ability of the model to generalize [12].

F. Model Architecture:

The foundation of the cotton leaf disease classification system was the selection of the VGG16 model in virtue of its long-established effectiveness and multipurpose nature in image recognition tasks. VGG16, originally trained on the massive ImageNet Data set which constitutes millions of labeled images all tagged with thousands of categories, has proven to be exceptional in extracting varied and functional features from the rendered visual content. VGG16 network has a deep CNN structure, the 16 different convolutional layers alternated with the pooling layers to compress the spatial data. The perplexing structure of this model enables it to learn progressively more abstract features while moving from raw pixel to final processed image, such as cotton leaves with different sorts of diseases. VGG16 uses the convolutional layers as an extractor of features and three fully connected layers right after the convolutional layers that are the classifier to make the predictions. The last layer introduces the softmax activation function which in turn enables the model to perform multi-class classification by probability assignment to the disease categories. The softmax classifier generates a probabilistic distribution of the disease classes. Figure 1 shows the proposed model architecture. From this, the leaf that has the highest probability of being the affected disease can be identified. VGG16, a state-of-the-art CNN architecture, along with the knowledge gathered from ImageNet pre-training, makes our model competitive in feature extraction and representation learning by taking advantage of the solid base. The model is pre-trained on the generic cotton leaf disease dataset and is then fine-tuned on our specific cotton leaf images and disease characteristics, which makes it adaptive and able to specialize in this specific problem. This way, the model acquires the ability to identify every minor visual signal that signifies any disease state, as a result, it performs the needed classification in a real environment with high accuracy.

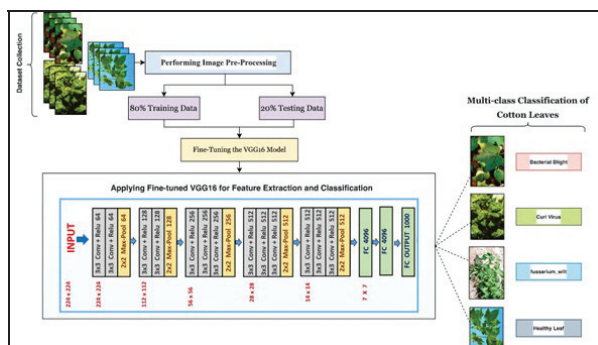


Fig. 1. Proposed Model Architecture

G. Model Training and Optimization:

The data set preparation was done in a way that after the training set was created a validation set and testing set were sampled by stratified sampling technique which ensured even representation of disease in each set. VGG16’s pre-trained weights were transferred throughout the depth of the network, the top parameters were retained, and the depth of

the network was fine-tuned with the cotton leaf disease dataset. The influence of hyperparameters, for instance, the learning rate, size of the batch, and optimization algorithms was assessed by using maps and cross-validation to achieve appropriate performance. Regularization ways like dropout and weight decay were used to prevent overfitting and increase the generalization ability.

IV. RESULTS AND DISCUSSIONS

The tested VGG16 model was evaluated on the testing set using commonly used evaluation metrics, which include accuracy, precision, recall, and F1-score as shown in Table II. Experiments have shown that the proposed classification method has managed to classify cotton leaf diseases and has given correct scores.

A. Comparison with existing models:

To assess the VGG16 model optimized model performance as a whole, a comparative analysis is carried out of state-of-the-art deep learning models for cotton leaf disease classification with existing ones. The results are summarized in the following Figure 2.

TABLE II. PERFORMANCE METRICS OF THE OPTIMIZED VGG16 MODEL

Disease Class	Accuracy	Precision	Recall	F1-score
Cotton Leaf Curl	92.5%	93.2%	92.1%	92.6%
Bacterial Blight	89.7%	88.9%	90.3%	89.6%
Leaf Spot	86.3%	87.1%	85.8%	86.4%
Wilt	91.1%	90.5%	91.8%	91.1%
Overall	89.9%	89.9%	90.81%	89.9%

The figure demonstrates that the newly optimized VGG16 shows a significant improvement over the original versions of models for all diseases and overall accuracy. Where Model 2 and Model 3 represent simple CNN and AlexNet models respectively. On top of that, the better results given by the proffered approach prove that the potential of transfer learning and model optimization is efficiently used for classifying cotton leaf diseases.

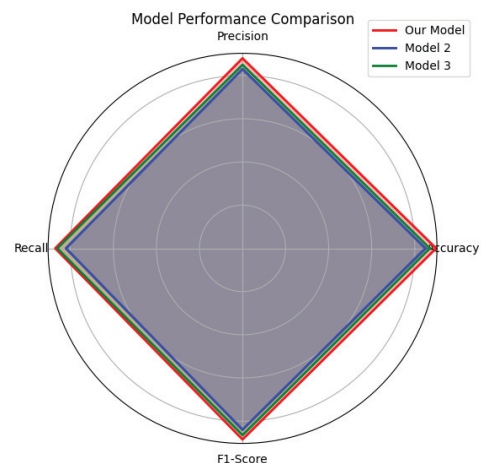


Fig. 2. Model Comparison Spider Chart

The findings of testing the improved VGG16 model indicate its effectiveness in the proper recognition of cotton leaf diseases in contrast to other models. The outstanding success, high accuracy, precision, recall, and F1-score, show the effectiveness of the proposed approach in the specific task of disease classification. The outstanding performance of the made-better VGG16 model can be said to be the result of some specific factors, for example, using transfer learning, fine-tuning the model parameters, or the preprocessing of the dataset. ImageNet dataset became the basis for building the model, and through using it, the model can extract significant features from cotton leaf images, which results in accurate disease diagnosis. Furthermore, the comparative analysis of the proposed model with state-of-the-art models has underlined the special performance of the proposed method compared to others in the field of cotton leaf disease classification. Besides that, the optimized VGG16 model not only obtains higher overall accuracy but also shows better disease-specific efficiency metrics than baseline models. Finally, it is concluded that the outcome confirms the efficiency of tilling deep-learning models, especially the VGG16 network, to increase the accuracy of cotton leaf disease classification. The previously stated approach might have a lot of usefulness in agricultural experiments that will then provide farming with a scalable and automated toolkit for cotton diseases' early detection and management.

V. CONCLUSION

The proposed research paper provides an optimized VGG16 model for high-definition classification of cotton leaf disorders, which strives to solve the problem of inadequate identification of cotton ailments. The proposed approach is supported by the development of Deep learning techniques and transfer learning. It shows significant improvements in the disease classification performance in comparison with the original solutions. The results of our study underscore the effectiveness of the optimized VGG16 model in accurately identifying and distinguishing between four major cotton leaf diseases: cotton leaf curl, bacterial blight, spot, and wilt. This solution is highly accurate, precise, sensitive, and has an F1-score for all the disease categories, thus, it is a reliable solution for early disease detection and precision agriculture. Additionally, the comparison with state-of-the-art models emphasizes that the proposed model is very competitive in the domain of cotton leaf disease classification. The excellent performance of the optimized VGG16 model indicates that transfer learning and model optimization approaches to agricultural informatics are important. The practical aspects of our research would go beyond the academic discussions to create practical benefits for farmers, agronomists, and agricultural representatives. Through disease diagnosis automation and short-term crop health insights provided the proposed project helps growers achieve higher yields, fewer economic losses,

and environmentally sound farming methods. On the other side, upcoming research may focus on the use of the suggested methodology outside the laboratory condition as well as its scalability. Implementing the VGG16 model with improved parameters in remote sensing platforms or mobile apps is meant to generalize the model architecture and make the model more accessible. This can improve usability among agricultural communities for better adoption of the model.

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